Mad Scientist 1

Four apparatus investigations for COP greater than one

Hob Nilre & Bo C. Herlin

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Abstract

Four open problems pursue an ambitious goal: a repeatable machine that delivers more useful energy than all the energy supplied to it. A falling-weight lever, switched transformer, contact chain and coupled-coil release each receive one page explaining the background, motivating observation and physical idea behind the question. Brief experiments point toward the next discovery; a shared energy account connects that discovery to complete-cycle performance.

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Local draft; the GitHub destination below is planned, not published.
Latest on GitHub: https://github.com/hobnilre/physics-mad-scientist-1

The top four

I want an apparatus that raises a load or charges a receiver, does it again, and leaves an energy record another experimenter can check. The ambition is a coefficient of performance (COP) greater than one: useful receiver energy divided by **all gross inward energy** over the complete history. Give the exciting output a receiver, give every input a meter, and see what survives.

This list alternates the two leading mechanical and two leading electrical investigations, preserving their order within each domain. The first pair starts the work; the second reuses much of its instrumentation. Rank sets experimental priority. The COP of each apparatus is a quantity to measure.

1. **The falling-weight lever.** Can compliant launch deliver a receiver gain that survives measured preparation and reset? [OP-CT-HW-01]
2. **The switched transformer.** Where does the work go when a winding connection is physically changed under load? [OP-TRF-02]
3. **The contact chain.** How do unequal masses, gaps and branching change loaded transfer when every mass is tracked? [OP-LR54-03]
4. **The coupled-coil release.** How is transfer divided among receiver, clamp and parasitic stores when a coupled circuit opens? [OP-LR48-03]

The starting points are striking: a fast launch, different voltages across different terminal pairs, and growing sums of momentum or flux-linkage magnitudes. Each suggests a physical idea worth following. Their open questions concern the contact histories and changing connections that turn the idea into an apparatus.

Each page traces that background and explains what made the question open, ending with a short route to an experiment. Demonstrating repeatable full-cycle excess output is the challenge. The final page supplies the common energy account.

1 The falling-weight lever

Background. A teeterboard turns the downward arrival of one or two people into the upward flight of another. It is a vivid starting point for a mad scientist: a short contact produces a dramatic launch, and changes in timing, leg compression or board flexibility can change the outcome. The falling-weight lever brings that idea to the tabletop, replacing the people with guided masses and a receiver while retaining the contact-transfer question.

The observation that starts the inquiry. Consider an illustrative comparison: two incoming masses, each $m = 70\,\mathrm{kg}$, move downward at $u = 5\,\mathrm{m/s}$, while one receiver of the same mass leaves upward at $w = 8\,\mathrm{m/s}$. Their incoming and outgoing kinetic energies differ by

$$\Delta K_{\mathrm{comparison}} = \frac{m}{2}(w^2 - 2u^2) = \frac{70}{2}(64 - 50)\,\mathrm{J} = 490\,\mathrm{J}. \tag{1}$$

That is an invitation to reconstruct the launch. The descending masses can continue losing gravitational potential energy during contact; elastic compression can store and return work; active leg forces can contribute along the way. These chosen endpoint speeds leave those histories to be identified.

Why this became an open problem. The launch also exchanges momentum with the support. An ideal fixed pivot can deliver an impulse while doing zero work, because its point of application stays still. Momentum delivered to the flyer and work delivered to the flyer therefore require different accounts. Board bending, leg stroke, catch engagement and the instant of separation make the route between the endpoints still richer.

Models with the same prescribed takeoff speed can have different force histories and active-work requirements. Watching the final flight alone leaves those possibilities together. The associated hardware question organizes the missing force, motion and calibration records needed to distinguish them. Takeoff time, board recoil, contact force and compression history are clues to the underlying mechanism.

The idea worth pursuing. Replace active performers with falling masses and let geometry, compliance and arrival timing do the organizing. A spring can receive work during one part of the motion and return it while the receiver moves differently. Two arrivals may interact through the board's ongoing rotation and deformation. The attractive question is how deliberately shaping that sequence changes useful delivery, including the work needed to prepare and restore the machine. This passive version is a proposed adaptation with its own model, separate from the human apparatus.

A first experiment. Compare a rigid and a compliant tabletop lever, then vary the delay between two falling masses. Track their motion and the receiver's force and displacement, including preparation and reset. One compact comparison can reveal whether compliance changes the transfer history in the way the idea requires.

2 The switched transformer

Background. A transformer winding with a tap offers several places to connect a receiver. The motivating analogy is a planet gear: its rotation relative to its carrier differs from its rotation relative to the ring. For the transformer, the corresponding choice is the pair of terminals across which voltage is observed. This makes the return connection an interesting part of the apparatus.

The observation that starts the inquiry. Call the terminals a, b, c. Winding 1 lies between a and c; winding 2 between b and a. On the same electrical trajectory, their voltage differences satisfy

$$V_b - V_c = (V_b - V_a) + (V_a - V_c). \tag{2}$$

For an ideal opposed winding ratio of -4 and an input $V_a - V_c = 1\,\mathrm{V}$, winding 2 gives $V_b - V_a = -4\,\mathrm{V}$ and the series connection gives $V_b - V_c = -3\,\mathrm{V}$. Both readings describe the same instant. The tempting idea is to exploit the two connections during different parts of a cycle.
A lightly loading meter reveals the two voltages. A receiver drawing current also changes the source's return path and the subsequent electrical motion. Each loaded connection consequently has its own current history. Shifting the numerical zero of every potential leaves the differences unchanged; moving the receiver's actual return wire creates the physical intervention.

Why this became an open problem. Comparing two fixed connections leaves a third piece of physics to supply: the transition between them. A winding may still carry current when its switch opens, and a capacitor may still hold voltage when a new connection is made. Current then follows a clamp, winding capacitance, an arc or another conducting path. Capacitors can redistribute charge. The selector and its driver participate in the same event.

The original fixed-connection descriptions retained current paths through their declared changes. They left actual opening and tap commutation to a fuller switching description. Later finite models provide examples of how that transition can be represented, while the real switch and gap laws remain to be identified. A bounded complete-history comparison required more supply in its switched case at matched receiver delivery, giving further designs a concrete benchmark.

The idea worth pursuing. Treat connection and timing as physical design variables. Can the receiver be connected where the evolving winding and capacitor states favor delivery, then reconnected for a different part of the cycle? How much of the subsequent transfer reaches the receiver, returns to the source or stays in the machine? The open question lives in this changing circuit and its real switching behavior.

A first experiment. Compare a fixed tap with a physically commutated tap on an isolated, current-limited low-voltage bench. Keep the receiver and clamp observable, and record synchronized source, driver and receiver power through switching and reset. This gives the proposed timing idea a direct comparison.

3 The contact chain

Background. A row of unequal masses joined by springs or compliant contacts can pass a disturbance from one end to the other. Some masses reverse while others keep advancing; a light terminal mass can acquire a large speed. The interesting possibility is to organize many local exchanges into a strong final delivery. Mass order, contact duration and the arrival sequence become design choices.

The observation that starts the inquiry. The underlying models track both the signed total momentum P and the sum of momentum magnitudes A_p. If p_j is the momentum of mass j along the line, then

$$A_p = \sum_j |p_j| = P + 2 \sum_{p_j<0} |p_j|, \qquad P = \sum_j p_j. \tag{3}$$

In a free chain, equal-and-opposite internal forces leave P unchanged. Equation (3) shows how A_p can nevertheless grow as backward and forward motion develop together. More motion in opposing directions can coexist with the same signed total.

That distinction is central to the original curiosity. Kinetic energy follows the separate expression $\sum_j p_j^2/(2m_j)$, with m_j the individual masses; compressed contacts carry elastic stores as well. The growing magnitude sum, the terminal speed and the energy available to a receiver are three different views of the evolving chain. Their relationship makes the system worth exploring.

Why this became an open problem. A spring joining two masses can push and pull continuously. A collision contact pushes during compression and releases when the bodies separate. Thus the set of interacting pairs changes during the motion. In the initial compliant descriptions the bodies start with zero gaps, the chain is free of fixtures, and the contact law sets a finite duration.

Actual gaps introduce travel between contacts and alter which events overlap. A branch offers more than one route for the impulse. Plastic deformation, restitution and fixture motion introduce further physical behavior. Making a contact increasingly stiff also raises a distinct question: what limiting impulse and work describe the event? These are the extensions that produced the selected OP. They connect a bounded compliant-chain description to richer physical contact networks.

The idea worth pursuing. Use the spacing and mass order to arrange when the terminal mass encounters the next transfer. A branch could change the timing or division of that transfer; a receiver feeds its reaction back into the chain. This turns the striking internal motion into a concrete design question: which combinations concentrate useful delivery, and what preparation and remaining stores accompany them?

A first experiment. Start with three unequal masses and spring bumpers. Compare touching and separated starts, then reverse the mass order. Track every mass and spring while a receiver takes work from the end. The full preparation and reset account connects the visible motion to the COP question.

4 The coupled-coil release

Background. Two inductors exchanging through a capacitor provide an electrical counterpart to the contact chain. Flux linkages can develop opposite signs while their signed sum stays fixed. The sum of their magnitudes can then grow. The provocative next idea is to change the connection at a chosen moment and direct part of that evolving state into a receiver.

The observation that starts the inquiry. A transparent ideal example uses the zero-mutual-coupling limit, $L_1 = 1\,\mathrm{H}$ and $L_2 = 19\,\mathrm{H}$, connected through a capacitor $C > 0$. Initially, $i_1 = 10\,\mathrm{A}$, $i_2 = 0$ and the capacitor voltage is zero. For the equal-and-opposite winding-voltage connection, the exact LC solution is

$$i_1 = \frac{1 + 19\cos\theta}{2}\,\mathrm{A}, \qquad i_2 = \frac{1 - \cos\theta}{2}\,\mathrm{A}, \tag{4}$$

where $\theta = \omega t$ and $\omega^2 = (1/L_1 + 1/L_2)/C$. At half an exchange cycle, $\theta = \pi$, the currents are -9 and $1\,\mathrm{A}$ and the capacitor voltage returns to zero. The flux linkages $\lambda_j = L_j i_j$ have moved from $(10, 0)$ to $(-9, 19)$ Wb. Their signed sum stays 10 Wb while their magnitude sum rises from 10 to 28 Wb.
The independently evaluated magnetic store is 50 J initially and $(81 + 19)/2 = 50\,\mathrm{J}$ at this endpoint. The example makes the motivating distinction visible: a growing flux-linkage magnitude and redistributed stored energy occur together. It supplies a precise starting state from which to ask what a different connection would deliver.

Why this became an open problem. Opening a branch changes the equations that linked the two winding voltages. The relation supporting the connected exchange must be reconsidered on the new circuit. Ideal zero-current opening offers a clean special case; finite-current opening brings the switch gap, clamp, winding capacitance and possible arc into the motion.

The real release therefore needs its own physical description. Which path carries the current? How does the clamp release? Can an arc reignite? What remains in the winding and capacitor states? These missing device laws and the complete circuit during release are the selected OP.

The idea worth pursuing. Choose the release time, coupling and winding orientation to steer the transfer into a receiver. With fixed reciprocal linear coupling, the magnetic store includes Mi_1i_2 in addition to $L_1 i_1^2/2 + L_2 i_2^2/2$, where M is the signed mutual inductance. The connected example is a starting comparison; the physical release asks a new question about the paths created by switching.

A first experiment. Prepare both coils through a metered source and release into a capacitor or resistor through an observable clamp. Compare zero-current and finite-current release, tracking both coil currents and receiver work. Include driver supplies, preparation and reset in the same account.

The same test on every bench

Draw the machine boundary before measuring. At every crossing port k, take power P_k positive inward: *vi* electrically, $\mathbf{F} \cdot \mathbf{v}$ mechanically, or $\tau \cdot \omega$ rotationally. Effort and motion refer to the same station and frame. Include signed heat and environmental transfers when present. For the complete interval $[a, b]$, define

$$W_k = \int_a^b P_k \, dt = D_k - X_k, \qquad D_k = \int_a^b \max(P_k, 0) \, dt, \qquad X_k = -\int_a^b \min(P_k, 0) \, dt, \tag{5}$$

$$E_{\text{in}} = \sum_k D_k, \qquad E_{\text{out}} = \sum_k X_k, \qquad r_E = \Delta U - E_{\text{in}} + E_{\text{out}}. \tag{6}$$

Here $\Delta U = U(b) - U(a)$ is evaluated independently from endpoint states; r_E is the signed balance residual. Positive r_E denotes outward excess relative to measured inputs and store change. Heat retained inside is a store change; heat crossing the boundary is a transfer. Count each once.

Let E_{useful} be outward work delivered to the declared useful receivers. Report source return and unwanted heat separately. With a resolved, positive input denominator, the ratio and its conservative measurement bound are

$$\text{COP} = \frac{E_{\text{useful}}}{E_{\text{in}}}, \qquad \text{COP}_{\text{lower}} = \frac{E_{\text{useful}} - B_{\text{out}}}{E_{\text{in}} + B_{\text{in}}}. \tag{7}$$

The nonnegative bounds B_{in} and B_{out} must have independently established simultaneous coverage. Declare their coverage and an input floor before evaluation; an unresolved or smaller input makes the ratio undefined. Identifying the ports and stores gives this bound its physical meaning.

Measure the whole history. Timestamp preparation, operation, switching/release, relaxation and actual reset, without gaps or double counting. Split integration at events and retain both event-side states. Measure finite transitions or separately justify any impulse work. Verify mechanical, electrical, magnetic, controller and thermal endpoint states as applicable. Independent store checks establish what a visible return means. Measure reset work directly and resolve possible depletion before assigning a full-cycle COP.

Calibrate and compare. Establish gains, offsets, timing skew, bandwidth and correlated errors with known loads or motions. Use simultaneous channels, independent receiver checks and event-resolved integration checks. Finer acquisition must resolve the actual events. Predeclare drive levels, receiver loads, independent preparations and controls; retain failed starts and both residual signs. Assess measurement, integration and model uncertainty separately.

Follow the interesting result. A lower COP bound above one, supported by the boundary and endpoint checks, earns independent repetition. An unexplained deficit also points to physics worth investigating. Keep exploratory adjustments separate from fresh confirmation measurements. Start with the lever and transformer, then carry the instruments into the contact chain and coil release. Build the strongest experiment, follow the unexpected result, and make every apparent win survive another meter and another independent preparation.